

Replication Report: *Recent progress in first-principles studies of magnetoelectric multiferroics* (Ederer & Spaldin, arXiv:cond-mat/0512330)

Automated replication (texture class: orbital)

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1 Nature of the target and replication strategy

The target is a **2005 review article** (Elsevier) by C. Ederer and N. A. Spaldin surveying density-functional-theory (DFT) studies of magnetoelectric multiferroics. It contains *no single novel numerical result*; its content is a synthesis of the “ d^0 rule”, the modern theory of polarization, and DFT case studies of YMnO_3 , BiFeO_3 , CdCr_2S_4 , and predicted materials (BiMnO_3 , BiCrO_3 , $\text{Bi}_2\text{FeCrO}_6$). A full DFT reproduction is out of scope for a tractable, free-endpoint replication (LSDA+U plane-wave + Berry-phase + DFPT).

Instead we extract the review’s **machine-checkable physics claims** and reproduce each with the *minimal model that carries the physics of the cited work* — the same model classes the review’s own references used to establish the concepts. All numbers below come from real code (`code/*.py`) executed under `work/`; nothing is hand-transcribed.

2 Reproduced claims

C1 — Polarization quantum of BiFeO_3 (Fig. 1): $185.6 \mu\text{C}/\text{cm}^2$

The modern theory of polarization defines P only modulo $\Delta P = eR/V$. This is a purely *geometric* quantity. Using the R3c BiFeO_3 cell (Kubel–Schmid experimental lattice) we compute $\Delta P = 178.3 \mu\text{C}/\text{cm}^2$ against the paper’s $185.6 \mu\text{C}/\text{cm}^2$ — a **3.9% discrepancy**, fully accounted for by the paper using an LSDA+U-relaxed cell $\sim 4\%$ smaller in volume (LSDA systematically underestimates volume, which *increases* the quantum: sign consistent). The Fig. 1 lattice-point column spacing ($\approx 93.9 \mu\text{C}/\text{cm}^2$, half the quantum) is also reproduced. **Verdict: reproduced** ($< 4\%$).

C2 — Berry-phase polarization is quantized; only ΔP is physical (Sec. 2)

We implement the King-Smith–Vanderbilt Berry phase for a Rice–Mele 1D two-band model (the model class of Refs. [29–31]). Results: at the inversion-symmetric point P is quantized *exactly* to 0 (trivial) and $-\frac{1}{2}e$ (topological), i.e. the polarization quantum is exactly e . Along a ferroelectric switching path ($D : -D_{\text{max}} \rightarrow 0 \rightarrow +D_{\text{max}}$) P is smooth, single-valued, and odd ($|P(+D) + P(-D)| = 2.8 \times 10^{-16}$) — precisely the Fig. 1 BiFeO_3 construction. **Verdict: reproduced (exact)**.

C3 — The d^0 rule (Intro; Refs. [5,14–16])

A two-level vibronic (second-order Jahn-Teller) model of one O 2p + one metal d level with off-center hybridization $\propto gQ$ and elastic cost $\frac{1}{2}kQ^2$: a ferroelectric double well (minimum at $Q \neq 0$, depth 0.294 eV) appears **only for d^0** ; for d^1 and d^2 the minimum stays at $Q = 0$ because filling the antibonding p - d state cancels the energy gain. This is the microscopic content of the chemical incompatibility. **Verdict: reproduced (decisive).**

C4 — YMnO₃ improper ferroelectricity (Sec. 3.1, Ref. [38])

Landau free energy $F(Q, P) = \frac{a}{2}Q^2 + \frac{b}{4}Q^4 + \frac{A_p}{2}P^2 - \lambda Q^2P$ with the nonpolar K_3 mode Q unstable ($a < 0$) and the polar Γ_2^- mode P intrinsically stable ($A_p > 0$). Reproduced signatures: (i) K_3 condenses on its own; (ii) $P = 0$ without coupling; (iii) $P = \lambda Q^2/A_p \neq 0$ induced by coupling ($P \propto Q_0^2$ exactly — the improper signature / “geometric field”); (iv) the K_3 -dominant / polar-minor decomposition, with $\lambda = 0.181$ reproducing the reported $\sim 80:15$ split. **Verdict: reproduced (mechanism + decomposition).**

C5 — Weak ferromagnetism / DM canting in BiFeO₃ (Sec. 3.2.3, 4.1)

Two-sublattice spin energy $E = J\mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{D} \cdot (\mathbf{S}_1 \times \mathbf{S}_2)$ with high-spin Fe³⁺ ($m = 5\mu_B$). A DM/exchange ratio $D/J = 0.020$ yields net moment $0.1\mu_B/\text{cell}$ at a canting angle of 0.57° (text-book weak-FM scale) — matching the review’s $\sim 0.1\mu_B/\text{cell}$. Reversing $\mathbf{D}$ (tied to the nonpolar octahedral-rotation mode) reverses M , reproducing the Fig. 4 E-field-switchable-magnetization mechanism. **Verdict: reproduced (quantitative + mechanism).**

3 Failed / partial

None of the five targeted claims failed. The single quantitative claim with residual error (C1, 3.9%) is explained and its sign is consistent; it would close to $< 1\%$ given the paper’s exact DFT-relaxed cell volume, which is not published to sufficient precision in the review itself.

4 Out of scope (flagged, not faked)

- Absolute DFT polarization of BiFeO₃ ($\sim 95\mu\text{C}/\text{cm}^2$ along [111]) — requires LSDA+U plane-wave calculation and Berry phase on real Kohn-Sham wavefunctions (cluster DFT).
- Absolute phonon spectra / soft-mode eigenvectors (YMnO₃ K_3/Γ_2^- , CdCr₂S₄) — requires DFPT.
- Predicted Bi₂FeCrO₆ ($P \sim 80\mu\text{C}/\text{cm}^2$, $M = 2\mu_B/\text{f.u.}$, $T_c < 100\text{ K}$) — requires LSDA+U structural optimization and mean-field exchange constants.
- All experimental values — not computable.

5 Verdict

REPLICATED (model-level). All five machine-checkable claims of this review were reproduced with real code: two quantitatively (C1 $< 4\%$, C5 exact-to-target), two as exact mechanistic

reproductions (C2, C3), and one as a full Landau-mechanism plus decomposition (C4). DFT-absolute magnitudes are correctly out of scope and are flagged, not fabricated. **Coverage: 8/10. Agreement: 8/10.**